MDCAT 2026 - SECOND-YEAR PRACTICE TEST

Sindh Textbook Board Focus | PMDC-Style MCQ Paper

Total MCQs: 180 **Time Allowed:** 3 Hours **Negative Marking:** None

Subject Weightage: Biology 81 | Chemistry 45 | Physics 36 | English 9 | Logical Reasoning 9

Difficulty Level: 27 Easy (15%) | 126 Moderate (70%) | 27 Difficult (15%)

Instructions: *Answer all questions. English contains active/passive voice, direct/indirect speech, subject-verb agreement, synonyms/antonyms, error detection and spelling. Logical Reasoning follows PMDC themes of critical thinking, letter/symbol series, logical deductions, logical problems, course of action, and cause/effect.*

BIOLOGY

1. Which hormone is mainly responsible for increasing water reabsorption in the kidneys?

A. ADH
B. Insulin
C. Thyroxine
D. Glucagon

2. The functional unit of the kidney is the:

A. Neuron
B. Nephron
C. Alveolus
D. Osteon

3. Which part of the brain helps maintain body temperature?

A. Medulla oblongata
B. Hypothalamus
C. Cerebellum
D. Pons

4. Insulin is produced by the:

A. Thyroid gland
B. Adrenal cortex
C. Pancreas
D. Pituitary gland

5. The hormone that increases blood glucose level is:

A. Insulin
B. Glucagon
C. ADH
D. Calcitonin

6. The basic contractile unit of a muscle fibre is the:

A. Sarcomere
B. Neuron
C. Tendon
D. Ligament

7. Which type of muscle is under voluntary control?

A. Cardiac
B. Smooth
C. Skeletal
D. Visceral

8. The male gamete in humans is called:

A. Ovum
B. Sperm
C. Zygote
D. Embryo

9. Fertilization in humans normally occurs in the:

A. Uterus
B. Ovary
C. Oviduct
D. Cervix

10. A patient produces very dilute urine despite dehydration. Which hormone is most likely deficient?

A. ADH
B. Insulin
C. Glucagon
D. Thyroxine

11. When blood glucose rises after a meal, the most immediate hormonal response is:

A. Increased glucagon release
B. Increased insulin release
C. Decreased insulin release
D. Increased ADH release

12. Which nephron region is especially important for establishing the medullary osmotic gradient?

A. Loop of Henle
B. Bowman's capsule
C. Glomerulus
D. Renal pelvis

13. The main role of the glomerulus is to:

A. Secrete urine into the bladder
B. Filter blood plasma
C. Reabsorb glucose only
D. Store urine

14. If the afferent arteriole diameter decreases, glomerular filtration rate will generally:

A. Increase
B. Decrease
C. Remain exactly unchanged
D. Become zero in all cases

15. A person with low blood calcium may increase secretion of:

A. Parathyroid hormone
B. Insulin
C. ADH
D. Growth hormone

16. Which sequence best represents a homeostatic control system?

A. Receptor → control centre → effector
B. Effector → receptor → stimulus
C. Control centre → stimulus → receptor
D. Stimulus → effector → receptor only

17. Damage to the cerebellum would most directly affect:

A. Balance and coordination
B. Hormone secretion only
C. Blood filtration
D. Hearing only

18. A withdrawal reflex can occur rapidly because its initial integration mainly occurs in the:

A. Spinal cord
B. Cerebrum
C. Cerebellum
D. Hypothalamus

19. The myelin sheath increases nerve impulse speed mainly by allowing:

A. Continuous slow conduction
B. Saltatory conduction
C. Chemical digestion
D. Hormone transport

20. At a synapse, neurotransmitter is released from the:

A. Postsynaptic membrane
B. Presynaptic terminal
C. Cell body only
D. Myelin sheath

21. If the sodium-potassium pump stops working for a long time, a neuron will eventually lose its:

A. Ability to maintain ion gradients
B. DNA
C. Cell membrane
D. Ribosomes

22. Which hormone promotes milk production after childbirth?

A. Prolactin
B. Oxytocin
C. FSH
D. LH

23. Oxytocin is especially important for:

A. Uterine contractions and milk ejection
B. Increasing blood glucose
C. Increasing calcium in bones
D. Producing sperm

24. FSH in females mainly stimulates:

A. Follicle development
B. Uterine contraction
C. Milk ejection
D. Ovulation directly

25. The LH surge in the menstrual cycle is most closely associated with:

A. Ovulation
B. Menstruation
C. Implantation
D. Menopause

26. During spermatogenesis, meiosis produces cells that are:

A. Diploid and genetically identical
B. Haploid and genetically varied
C. Triploid
D. Always genetically identical

27. The acrosome of a sperm contains enzymes that help it:

A. Produce ATP
B. Penetrate egg coverings
C. Replicate DNA
D. Form the placenta

28. Implantation normally occurs in the:

A. Endometrium
B. Ovary
C. Oviduct wall
D. Cervix

29. The placenta functions mainly as a site for:

A. Exchange between maternal and fetal blood systems
B. Direct mixing of all maternal and fetal blood
C. Sperm production
D. Bile storage

30. Which structure prevents entry of additional sperm after fertilization?

A. Changes in the egg coverings
B. Seminal vesicle
C. Prostate gland
D. Urethra

31. During embryonic development, the ectoderm gives rise mainly to:

A. Nervous system and epidermis
B. Muscles and bones only
C. Digestive tract only
D. Blood plasma only

32. DNA replication is described as semiconservative because:

A. Both new strands are completely new
B. Each daughter DNA has one old and one new strand
C. DNA is copied only once
D. RNA replaces DNA

33. If a DNA template strand is TAC-GGA, the mRNA sequence is:

A. ATG-CCT
B. AUG-CCU
C. UAC-GGA
D. TAC-GGA

34. A mutation changes a codon into another codon for the same amino acid. This is a:

A. Silent mutation
B. Nonsense mutation
C. Frameshift mutation
D. Chromosomal deletion

35. A mutation that introduces a premature stop codon is called:

A. Silent
B. Missense
C. Nonsense
D. Neutral

36. Crossing over occurs during:

A. Prophase I of meiosis
B. Prophase of mitosis
C. Telophase II only
D. Interphase only

37. Independent assortment is mainly due to the:

A. Random orientation of homologous chromosome pairs
B. Replication of DNA
C. Formation of ribosomes
D. Fusion of gametes

38. A carrier female for an X-linked recessive disorder marries a normal male. What proportion of their sons is expected to be affected?

A. 0%
B. 25%
C. 50%
D. 100%

39. In a dihybrid cross AaBb x AaBb, the classic phenotypic ratio under independent assortment is:

A. 3:1
B. 1:1
C. 9:3:3:1
D. 1:2:1

40. A person with genotype I^AI^B has blood group:

A. A
B. B
C. AB
D. O

41. A person with blood group O has genotype:

A. I^AI^A
B. I^BI^B
C. I^AI^B
D. ii

42. Which event increases variation without creating a new DNA sequence by itself?

A. Independent assortment
B. DNA replication
C. Mitosis
D. Binary fission

43. A plasmid used to carry a desired gene into bacteria is called a:

A. Vector
B. Antibody
C. Ribosome
D. Codon

44. Restriction enzymes are useful in genetic engineering because they:

A. Cut DNA at specific sequences
B. Join amino acids
C. Produce ATP
D. Translate mRNA

45. DNA ligase is used to:

A. Join DNA fragments
B. Cut DNA
C. Make RNA from DNA
D. Destroy proteins

46. A genetically modified bacterium is commonly used to produce human:

A. Insulin
B. Hemoglobin directly from air
C. Bone tissue
D. Bile

47. A species becomes reproductively isolated from another population. This can promote:

A. Speciation
B. Digestion
C. Homeostasis
D. Translation

48. Fossils are important evidence for evolution because they can show:

A. Changes in organisms through geological time
B. Only present-day mutations
C. Only food chains
D. Only cell division

49. In a food chain, energy available generally:

A. Increases at higher trophic levels
B. Decreases at higher trophic levels
C. Remains exactly the same
D. Becomes zero at producers

50. The accumulation of a toxic substance at higher trophic levels is called:

A. Biomagnification
B. Nitrogen fixation
C. Transpiration
D. Succession

51. Nitrogen-fixing bacteria help convert atmospheric nitrogen into:

A. Usable nitrogen compounds
B. Oxygen
C. Glucose directly
D. Carbon dioxide

52. A population that exceeds the carrying capacity of its environment is likely to experience:

A. Greater resource pressure
B. Unlimited growth forever
C. No competition
D. No mortality

53. Primary succession begins in an area that:

A. Lacks developed soil
B. Has a mature forest
C. Has a stable climax community
D. Has abundant farmland

54. The greenhouse effect is mainly associated with increased atmospheric:

A. Heat-trapping gases
B. Nitrogen only
C. Oxygen only
D. Water-free air

55. Which is a renewable resource?

A. Solar energy
B. Coal
C. Natural gas
D. Petroleum

56. A decrease in predator numbers may cause prey numbers to:

A. Increase, at least initially
B. Always decrease
C. Become zero
D. Remain mathematically fixed

57. Which process directly returns carbon dioxide to the atmosphere?

A. Respiration
B. Photosynthesis
C. Nitrogen fixation
D. Protein synthesis

58. A population bottleneck can reduce genetic diversity because:

A. A large population is suddenly reduced to a small number
B. Every gene mutates
C. All individuals become identical immediately
D. Natural selection stops

59. Which statement best describes genetic drift?

A. Random change in allele frequencies, especially in small populations
B. Directed change caused only by need
C. Production of ATP
D. Exchange of gases

60. The Hardy-Weinberg principle assumes, among other conditions:

A. Random mating and no natural selection
B. Strong natural selection
C. Very small population
D. Continuous mutation at high rate

61. If p = 0.7 and q = 0.3, the expected frequency of heterozygotes under Hardy-Weinberg equilibrium is:

A. 0.21
B. 0.42
C. 0.49
D. 0.91

62. A frameshift mutation is most likely caused by:

A. Insertion or deletion of bases not in multiples of three
B. A substitution that changes no amino acid
C. Crossing over
D. Independent assortment

63. Which molecule carries amino acids to the ribosome during translation?

A. tRNA
B. DNA
C. rRNA only
D. ATP

64. The anticodon is found on:

A. tRNA
B. DNA
C. mRNA
D. Protein

65. A stop codon normally signals:

A. Termination of translation
B. Start of DNA replication
C. Start of transcription
D. Cell division

66. Which statement about homologous chromosomes is correct?

A. They carry genes for the same traits at corresponding loci
B. They are always identical copies
C. They occur only in gametes
D. They contain no DNA

67. The cell cycle checkpoint that prevents a cell with severe DNA damage from entering division mainly helps:

A. Maintain genomic stability
B. Increase mutation rate
C. Stop all protein synthesis
D. Produce gametes directly

68. Meiosis is important in sexual reproduction because it:

A. Reduces chromosome number and generates variation
B. Doubles chromosome number in gametes
C. Produces genetically identical diploid cells
D. Stops crossing over

69. If a diploid organism has 24 chromosomes, its normal gametes contain:

A. 6
B. 12
C. 24
D. 48

70. A dominant phenotype does not always mean an individual is:

A. Homozygous dominant
B. Heterozygous
C. Affected by a dominant allele
D. Showing the dominant trait

71. Which biotechnology method can identify a person by comparing DNA patterns?

A. DNA profiling
B. Photosynthesis
C. Dialysis
D. Vaccination

72. A clone produced from a single parent is expected to be:

A. Genetically very similar to the parent
B. Always genetically unrelated
C. Always haploid
D. Unable to reproduce

73. A heterozygous carrier female for an X-linked recessive disease has children with an affected male. What is the probability that a randomly selected daughter is affected?

A. 0%
B. 25%
C. 50%
D. 100%

74. In a population, $q^2 = 0.09$ for a recessive allele under Hardy-Weinberg equilibrium. What is the carrier frequency?

A. 0.09
B. 0.21
C. 0.42
D. 0.91

75. A gene mutation changes a coding sequence by inserting two nucleotides near its beginning. The most likely result is:

A. A frameshift affecting many downstream codons
B. No effect on the reading frame
C. Only one amino acid substitution in all cases
D. Immediate chromosome duplication

76. A nephron is exposed to a drug that blocks the Na-K-2Cl cotransporter in the ascending limb. Which effect is most likely?

A. Reduced medullary osmotic gradient and reduced urine-concentrating ability
B. Increased glucose reabsorption
C. Increased ADH synthesis
D. Complete prevention of filtration

77. If the LH receptor on ovarian follicle cells is nonfunctional, which event is most directly impaired?

A. Ovulation and corpus luteum formation
B. DNA replication in every cell
C. Milk production only
D. Implantation only

78. Two genes are very close together on the same chromosome. Compared with genes that assort independently, their offspring are expected to show:

A. More parental combinations and fewer recombinant combinations
B. Only recombinant combinations
C. No inheritance pattern
D. Equal recombinant and parental combinations in every cross

79. A population has allele frequencies p = 0.8 and q = 0.2. After selection, the q allele becomes 0.1. The best interpretation is:

A. The allele frequency changed because evolutionary forces acted on the population
B. Hardy-Weinberg equilibrium was perfectly maintained
C. No evolution occurred
D. The population must have become extinct

80. A DNA sample contains 30% adenine. In double-stranded DNA, the approximate percentage of cytosine is:

A. 20%
B. 30%
C. 35%
D. 40%

81. If an ecosystem loses most of its decomposers, the most direct long-term consequence is:

A. Slower nutrient recycling and accumulation of dead organic matter
B. Immediate increase in primary production
C. Complete removal of carbon dioxide
D. Permanent increase in energy available to consumers

CHEMISTRY

82. Which functional group is characteristic of alcohols?

A. -OH
B. -COOH
C. -CHO
D. -CO-

83. The general formula of an alkene is:

A. C_nH_{2n}
B. C_nH_{2n+2}
C. C_nH_{2n-2}
D. C_nH_n

84. The functional group of a carboxylic acid is:

A. -COOH
B. -OH
C. $-NH_2$
D. -CHO

85. Benzene is an example of a/an:

A. Aromatic hydrocarbon
B. Alcohol
C. Carboxylic acid
D. Alkyl halide

86. The monomer of polyethylene is:

A. Ethene
B. Ethyne
C. Methane
D. Benzene

87. Which element is present in proteins but not normally in carbohydrates?

A. Nitrogen
B. Carbon
C. Hydrogen
D. Oxygen

88. The polymer formed from amino acids is a:

A. Protein
B. Lipid
C. Nucleic acid only
D. Carbohydrate

89. The major product of addition of HBr to propene in the absence of peroxide is:

A. 1-bromopropane
B. 2-bromopropane
C. 1,2-dibromopropane
D. Propanol

90. Oxidation of a primary alcohol under controlled conditions commonly gives a/an:

A. Aldehyde
B. Ketone
C. Ether
D. Alkane

91. Oxidation of a secondary alcohol generally gives a:

A. Ketone
B. Aldehyde
C. Carboxylic acid directly
D. Alkene only

92. Which reagent can distinguish an aldehyde from a ketone?

A. Tollens' reagent
B. Sodium chloride
C. Conc. sulfuric acid only
D. Bromine water only

93. A positive Tollens' test usually produces:

A. Silver mirror
B. Blue gas
C. Green precipitate
D. Black smoke

94. Ethanoic acid reacts with ethanol in the presence of acid to form:

A. Ethyl ethanoate
B. Methanal
C. Ethene
D. Ethanamide

95. Esterification is a reaction between a carboxylic acid and an:

A. Alcohol
B. Alkene
C. Amine only
D. Alkane

96. The IUPAC name of CH_3CH_2OH is:

A. Ethanol
B. Methanol
C. Propanol
D. Ethanal

97. The IUPAC name of CH_3COOH is:

A. Ethanoic acid
B. Methanoic acid
C. Propanone
D. Ethanol

98. Which hydrocarbon gives a sooty flame because of its relatively high carbon content?

A. Benzene
B. Methane
C. Ethane
D. Propane

99. The product of complete combustion of a hydrocarbon contains:

A. CO_2 and H_2O
B. CO and H_2
C. C and H_2
D. Only CO

100. Chlorination of methane in sunlight is an example of:

A. Free-radical substitution
B. Electrophilic addition
C. Nucleophilic substitution only
D. Esterification

101. Bromine water is decolorized by many alkenes because of:

A. Addition across the C=C bond
B. Substitution of all H atoms
C. Oxidation to CO_2 only
D. Formation of a protein

102. The carbon atoms in benzene are approximately:

A. sp^2 hybridized
B. sp hybridized only
C. sp^3 hybridized only
D. Unhybridized

103. Which compound can form hydrogen bonds strongly with water?

A. Ethanol
B. Ethene
C. Benzene
D. Ethane

104. Phenol is more acidic than ethanol mainly because:

A. Its conjugate base is resonance-stabilized
B. It has more carbon atoms
C. It is an alkene
D. It has no oxygen

105. When phenol reacts with bromine water, it commonly forms:

A. 2,4,6-tribromophenol
B. Bromomethane
C. Benzoic acid only
D. Ethene

106. Aldehydes contain the group:

A. -CHO
B. -COOH
C. -OH only
D. -COO-

107. Ketones contain a carbonyl group:

A. Between two carbon atoms
B. At the end of a carbon chain with H
C. Only in carboxylic acids
D. Attached directly to nitrogen only

108. Which compound is a ketone?

A. Propanone
B. Propanal
C. Propanol
D. Propanoic acid

109. Reduction of an aldehyde generally gives a:

A. Primary alcohol
B. Secondary alcohol
C. Carboxylic acid
D. Alkene

110. Reduction of a ketone generally gives a:

A. Secondary alcohol
B. Primary alcohol
C. Carboxylic acid
D. Aldehyde

111. Carboxylic acids react with sodium bicarbonate to release:

A. CO_2
B. H_2
C. O_2
D. N_2

112. Which is a strong electron-withdrawing effect that can increase acidity of a nearby carboxylic acid?

A. Inductive effect of electronegative atoms
B. Hydrogen bonding only
C. Polymerization
D. Combustion

113. The peptide bond forms between:

A. Amino and carboxyl groups
B. Two hydroxyl groups
C. Two phosphate groups
D. Two hydrocarbon groups

114. Denaturation of a protein usually affects its:

A. Three-dimensional structure
B. Amino acid sequence in every case
C. Atomic numbers
D. Peptide bond count in every case

115. Enzymes speed up reactions mainly by:

A. Lowering activation energy
B. Increasing the reaction's equilibrium constant
C. Being permanently consumed
D. Increasing product energy

116. The active site of an enzyme is the region that:

A. Binds the substrate and participates in catalysis
B. Stores DNA
C. Produces oxygen in all cells
D. Always contains DNA

117. Addition polymers are formed when monomers:

A. Add together without eliminating a small molecule
B. Always release CO_2
C. Always form water
D. Must contain amino acids

118. Propene reacts with HBr in the presence of organic peroxide. The major product is:

A. 1-bromopropane
B. 2-bromopropane
C. 1,2-dibromopropane
D. Propanone

119. Which pair correctly represents an oxidation sequence for a primary alcohol under sufficiently strong oxidation conditions?

A. Primary alcohol → aldehyde → carboxylic acid
B. Primary alcohol → ketone → alkane
C. Primary alcohol → alkene → ketone only
D. Primary alcohol → ester → alkane

120. A compound gives a positive Tollens' test and on oxidation forms ethanoic acid. The original compound is:

A. Ethanal
B. Propanone
C. Ethanol
D. Ethene

121. Which statement best explains the higher acidity of phenol compared with ethanol?

A. Phenoxide ion is resonance-stabilized while ethoxide ion is not
B. Phenol contains no oxygen
C. Ethanol is aromatic
D. Phenol has a higher molecular mass only

122. If 0.20 mol ethanoic acid reacts with 0.10 mol ethanol in an esterification reaction with excess acid catalyst, the maximum theoretical moles of ester are:

A. 0.05
B. 0.10
C. 0.20
D. 0.30

123. Which monomer pair can form a condensation polymer such as nylon?

A. A diamine and a dicarboxylic acid
B. Ethene only
C. Benzene and methane
D. Methane and chlorine

124. An alkene produces 2-bromopropane as the major product on reaction with HBr. The most likely alkene is:

A. Propene
B. Ethene
C. But-1-ene only
D. Propane

125. A carbonyl compound does not react with Tollens' reagent but is reduced to propan-2-ol. It is most likely:

A. Propanone
B. Propanal
C. Propanoic acid
D. Propanol

126. A reaction converts CH_3COOH into $CH_3COOCH_2CH_3$. The reaction is best classified as:

A. Esterification
B. Hydrogenation
C. Free-radical substitution
D. Polymer cracking

PHYSICS

127. The SI unit of electric charge is:

A. Coulomb
B. Volt
C. Ohm
D. Tesla

128. The SI unit of electric potential is:

A. Volt
B. Coulomb
C. Ampere
D. Farad

129. Ohm's law is:

A. V = IR
B. P = VI only
C. Q = It only
D. F = qE only

130. The SI unit of resistance is:

A. Ohm
B. Volt
C. Ampere
D. Weber

131. A transformer works on the principle of:

A. Electromagnetic induction
B. Electrolysis
C. Photoelectric effect
D. Nuclear fission

132. A diode mainly allows current to flow:

A. Preferentially in one direction
B. Equally in both directions
C. Only through vacuum
D. Only at zero voltage

133. Two charges of equal magnitude and opposite sign are placed close together. The electric field at the midpoint is directed:

A. From positive charge toward negative charge
B. From negative toward positive
C. Perpendicular to the line always
D. Zero in all cases

134. If the distance between two point charges is doubled, the electrostatic force becomes:

A. One-fourth
B. One-half
C. Double
D. Four times

135. A capacitor stores energy mainly in its:

A. Electric field
B. Magnetic field only
C. Mechanical motion
D. Nuclear field

136. When a capacitor is fully charged in a DC circuit, the ideal capacitor behaves as:

A. An open circuit
B. A short circuit
C. A battery with zero voltage
D. A resistor of fixed 1 Ω

137. A 6 Ω resistor connected to 12 V draws a current of:

A. 0.5 A
B. 2 A
C. 6 A
D. 72 A

138. If two identical resistors are connected in parallel, their equivalent resistance is:

A. Half the resistance of one resistor
B. Double one resistor
C. Equal to one resistor
D. Four times one resistor

139. The resistivity of a material mainly depends on:

A. Material and temperature
B. Length only
C. Area only
D. Current only

140. A source has emf 12 V and internal resistance 2 Ω. If the external resistance is 4 Ω, the current is:

A. 1 A
B. 2 A
C. 3 A
D. 6 A

141. Magnetic force on a moving charge is maximum when the charge moves:

A. Perpendicular to the magnetic field
B. Parallel to the field
C. At rest
D. Only at 0°

142. The force on a current-carrying conductor in a magnetic field is maximum when the conductor is:

A. Perpendicular to the field
B. Parallel to the field
C. At rest
D. Outside the field

143. Magnetic flux through an area is maximum when the magnetic field is:

A. Perpendicular to the surface
B. Parallel to the surface
C. Zero
D. At 90° to the area vector

144. Lenz's law gives the direction of induced current so that it:

A. Opposes the change producing it
B. Always increases the change
C. Has no relation to energy
D. Always flows clockwise

145. Increasing the number of turns in a coil generally increases the magnitude of induced emf because:

A. Flux linkage changes more strongly
B. Resistance becomes zero
C. Charge disappears
D. The magnetic field becomes impossible

146. A step-up transformer has:

A. More turns on the secondary than the primary
B. Fewer secondary turns
C. Equal turns only
D. No secondary coil

147. In an ideal transformer, if voltage is increased, current on the secondary side generally:

A. Decreases for the same power
B. Increases without limit
C. Remains exactly equal
D. Becomes zero

148. A PN junction diode is forward biased when:

A. P-side is connected to positive and N-side to negative
B. P-side is negative and N-side positive
C. Both sides are positive
D. Both sides are negative

149. A half-wave rectifier uses a diode to convert:

A. AC into pulsating DC
B. DC into AC only
C. Heat into electricity
D. Light into sound

150. In the photoelectric effect, increasing light frequency above threshold generally increases the:

A. Maximum kinetic energy of emitted electrons
B. Number of protons
C. Rest mass of photons
D. Charge of electrons

151. The energy of a photon is given by:

A. $E = hf$
B. $E = mc^2$ only
C. $E = IR$
D. $E = q/t$

152. An emission line spectrum is produced when electrons:

A. Move from higher to lower energy levels
B. Always remain in the same level
C. Leave the nucleus
D. Become neutrons

153. The half-life of a radioactive substance is the time required for:

A. Half the undecayed nuclei to decay
B. All nuclei to decay
C. The mass to double
D. The activity to become infinite

154. A 4 μF capacitor is connected to 100 V. The energy stored is:

A. 0.002 J
B. 0.02 J
C. 0.2 J
D. 2 J

155. Two capacitors of 2 μF and 4 μF are connected in series. Their equivalent capacitance is:

A. 1.33 μF
B. 2 μF
C. 6 μF
D. 8 μF

156. A charged particle enters a uniform magnetic field perpendicular to its velocity. Its path is:

A. Circular
B. Straight with constant velocity
C. Parabolic
D. Random

157. An electron and a proton enter the same magnetic field with equal speeds perpendicular to the field. Which has the larger circular radius?

A. Proton
B. Electron
C. Both equal
D. Cannot be determined because charge is absent

158. A transformer has 500 primary turns and 2500 secondary turns. If the primary voltage is 20 V, the ideal secondary voltage is:

A. 4 V
B. 20 V
C. 100 V
D. 2500 V

159. A radioactive sample has a half-life of 4 h. What fraction remains after 12 h?

A. 1/2
B. 1/4
C. 1/8
D. 1/16

160. An induced current opposes the change in magnetic flux. If the external magnetic flux into the page is increasing, the induced magnetic field is directed:

A. Out of the page
B. Into the page
C. Parallel to the page
D. Zero

161. A metal has a threshold frequency f. Light of frequency 2f falls on it. If the frequency is increased to 3f, the maximum kinetic energy changes by:

A. hf
B. 2hf
C. 3hf
D. hf/2

162. A resistor of 6 Ω and an inductor are connected to an AC source. If the inductive reactance equals 8 Ω, the impedance magnitude is:

A. 2 Ω
B. 8 Ω
C. 10 Ω
D. 14 Ω

ENGLISH

163. Change into passive voice: 'The committee approved the proposal.'

A. The proposal was approved by the committee.
B. The proposal is approved by the committee.
C. The committee was approved by the proposal.
D. The proposal had approved the committee.

164. Change into active voice: 'The patient was examined by the doctor.'

A. The doctor examined the patient.
B. The doctor was examined by the patient.
C. The patient examined the doctor.
D. The doctor has been examined by the patient.

165. Change into indirect speech: He said, 'I am tired.'

A. He said that he was tired.
B. He said that I am tired.
C. He says that he was tired.
D. He said that he is tired.

166. Change into indirect speech: Sara said, 'I have finished my work.'

A. Sara said that she had finished her work.
B. Sara said that I have finished my work.
C. Sara said that she has finished her work.
D. Sara says that she had finished my work.

167. Choose the correct sentence.

A. Neither the teacher nor the students are ready.
B. Neither the teacher nor the students is ready.
C. Neither the teacher or the students are ready.
D. Neither the teacher nor the students was ready.

168. Choose the synonym of 'abundant'.

A. Plentiful
B. Rare
C. Weak
D. Empty

169. Choose the antonym of 'scarce'.

A. Abundant
B. Limited
C. Rare
D. Insufficient

170. Identify the incorrect word: 'He recieved the message yesterday.'

A. He
B. recieved
C. message
D. yesterday

171. Choose the correctly spelled word.

A. Accommodation
B. Acommodation
C. Accomodation
D. Accommadation

LOGICAL REASONING

172. Series: 2, 5, 10, 17, 26, ?

A. 35
B. 37
C. 39
D. 41

173. Series: A, C, F, J, O, ?

A. T
B. U
C. V
D. W

174. All surgeons are doctors. Some doctors are researchers. Which conclusion is definitely true?

A. All surgeons are doctors.
B. All doctors are surgeons.
C. Some researchers are surgeons.
D. No doctor is a researcher.

175. Statement: 'The city should add more buses because many commuters are unable to find seats.' Which assumption is most reasonable?

A. More buses could help reduce overcrowding.
B. All commuters dislike buses.
C. Cars should be banned immediately.
D. Every bus is currently empty.

176. Five students A, B, C, D and E stand in a line. A is before B, C is after D, and E is at the end. Which arrangement is possible?

A. A-B-D-C-E
B. B-A-D-C-E
C. D-C-B-A-E
D. A-D-C-E-B

177. A shop's sales increased after it started online advertising. Which is the strongest conclusion?

A. Online advertising may have contributed to the increase.
B. Online advertising is the only possible cause.
C. Sales will always increase.
D. Customers no longer visit shops.

178. Course of action: A school notices that many students arrive late because the main road is frequently congested. Which action logically follows?

A. Adjust school arrival arrangements after studying traffic patterns.
B. Punish every late student without checking the cause.
C. Close the school permanently.
D. Ignore the problem.

179. Cause and effect: Heavy rainfall continued for several hours, and several low-lying roads became flooded. What is the most direct relationship?

A. The rainfall likely contributed to the flooding.
B. The flooding caused the rainfall.
C. The roads caused the clouds.
D. There is no possible relationship.

180. A box contains red, blue and green balls. Every red ball is larger than every blue ball, and every blue ball is larger than every green ball. If one ball is selected and it is not green, what can be concluded?

A. It is red or blue.
B. It must be red.
C. It must be green.
D. It is smaller than every green ball.

ANSWER KEY

Use this section only after completing the test.

Biology (Q1 - Q81)

1	A	2	B	3	B	4	C	5	B	6	A	7	C	8	B	9	C	10	A
11	B	12	A	13	B	14	B	15	A	16	A	17	A	18	A	19	B	20	B
21	A	22	A	23	A	24	A	25	A	26	B	27	B	28	A	29	A	30	A
31	A	32	B	33	B	34	A	35	C	36	A	37	A	38	C	39	C	40	C
41	D	42	A	43	A	44	A	45	A	46	A	47	A	48	A	49	B	50	A
51	A	52	A	53	A	54	A	55	A	56	A	57	A	58	A	59	A	60	A
61	B	62	A	63	A	64	A	65	A	66	A	67	A	68	A	69	B	70	A
71	A	72	A	73	C	74	C	75	A	76	A	77	A	78	A	79	A	80	A
81	A																		

Chemistry (Q82 - Q126)

82	A	83	A	84	A	85	A	86	A	87	A	88	A	89	B	90	A	91	A
92	A	93	A	94	A	95	A	96	A	97	A	98	A	99	A	100	A	101	A
102	A	103	A	104	A	105	A	106	A	107	A	108	A	109	A	110	A	111	A
112	A	113	A	114	A	115	A	116	A	117	A	118	A	119	A	120	A	121	A
122	B	123	A	124	A	125	A	126	A										

Physics (Q127 - Q162)

127	A	128	A	129	A	130	A	131	A	132	A	133	A	134	A	135	A	136	A
137	B	138	A	139	A	140	B	141	A	142	A	143	A	144	A	145	A	146	A
147	A	148	A	149	A	150	A	151	A	152	A	153	A	154	B	155	A	156	A
157	A	158	C	159	C	160	A	161	A	162	C								

English (Q163 - Q171) & Logical Reasoning (Q172 - Q180)

163	A	164	A	165	A	166	A	167	A	168	A	169	A	170	B	171	A	172	B
173	B	174	A	175	A	176	A	177	A	178	A	179	A	180	A				

> ***Pattern reference:*** *PMDC's 2026 public notice states 180 MCQs with Biology 81, Chemistry 45, Physics 36, English 9 and Logical Reasoning 9, with 15% Easy, 70% Moderate and 15% Difficult. The PMDC syllabus page currently lists the MDCAT 2025 curriculum; this paper uses that published curriculum for the syllabus areas while applying the 2026 test structure.*